

Lecture 7: Part I - Preferences

ECON 101
Fall 2020



A model of consumer preferences: the Questionnaire "Q"

- Long list of questions of the following type:

$Q(x, y) \div x, y$ are two distinct bundles:

How do you compare x to y ? Check one and only one of the following three options:

☐ I prefer x to y ($x \succ y$)

☐ I prefer y to x ($y \succ x$)

☐ I am indifferent ($x \sim y$)

A model of consumer preferences is just a long list of questions of this type and their answers.

What is a legal answer to the questionnaire?

- Restrictions (assumptions) on answers:

① is called
COMPLETENESS

① For any question $Q(x, y)$ exactly one of the boxes is checked.

We don't allow:

- Refraining to answer (no box ticked)
- Checking more than one box

All included!

This thus means that we exclude attitudes like:

- They are not comparable
- I don't know what x is
- I have no opinion
- I prefer x over y and y over x

② Suppose I have three bundles x, y, z such that

$$Q(x, y) : x \succ y$$

$$Q(y, z) : y \succ z$$

Then will imply that

$$Q(x, z) : x \succ z$$

$$Q(x, y) : x \sim y$$

$$Q(y, z) : y \sim z$$

Then this will imply that

$$Q(x, z) : x \sim z$$

② is called TRANSITIVITY

③ Since we are talking about goods, which are assumed to be desirable, consumer always prefer more of a good, i.e. the more the better.

For example: two bundles

$E = (30 \text{ food}, 40 \text{ clothing})$	} $Q(E, A)$
$A = (20 \text{ food}, 30 \text{ clothing})$	
	$\checkmark$ I prefer E to A
	$\nwarrow$
$E > A$	

③ is called "More-is-better" or MONOTONICITY

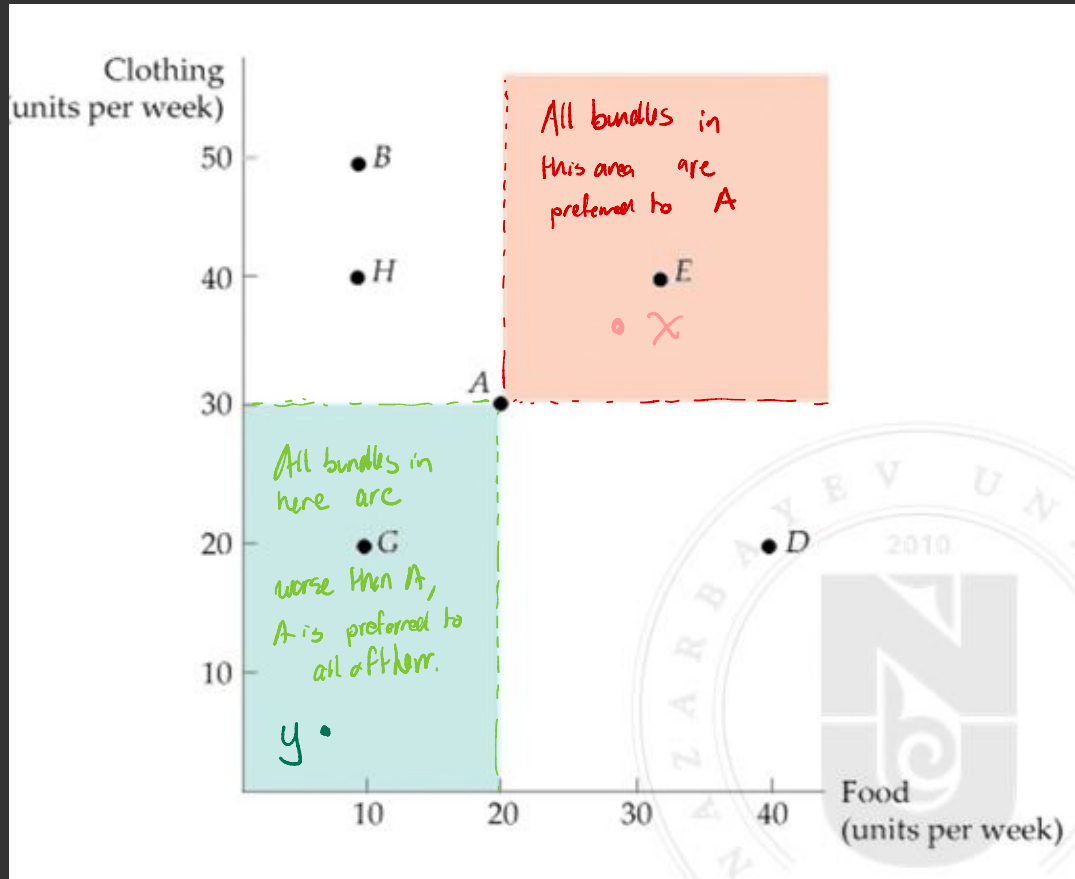
Under the three assumptions, of completeness, transitivity and more is better:
with respect to bundle A

iii A is preferred to any bundle in this area

iv Any bundle in the pink area is preferred to A

Ex: $A > y$

$x > A$



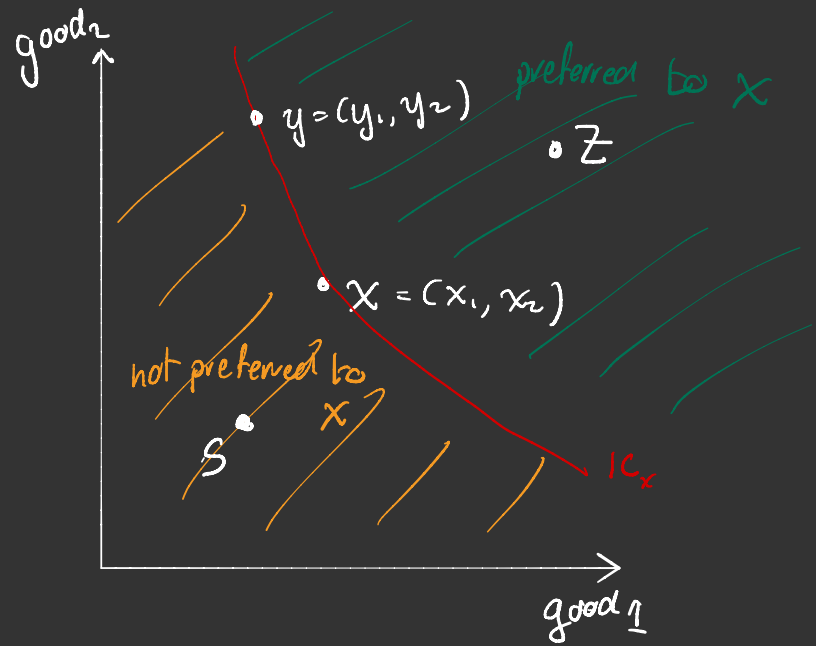
Indifference Curves

For a bundle x , the Indifference curve at x consists of all bundles indifferent to x .

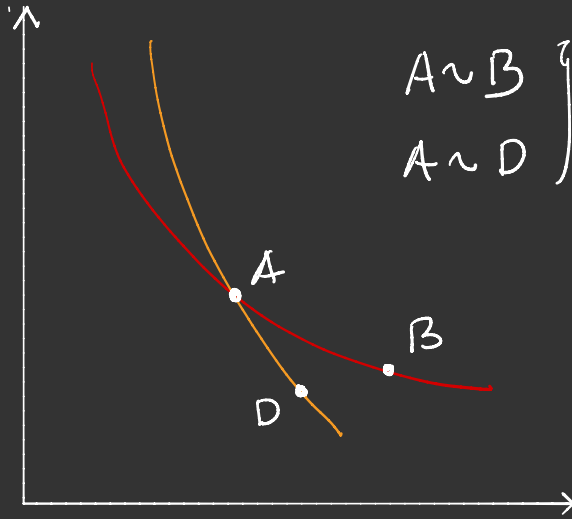
y is in IC through x ,
 $y \sim x$

Any bundle above IC_x , say
 $z > x$.

Any bundle below, IC_x , say
 $x \succ s$



- Completeness Implies that the indifference map is "full" of indifference curves.
- Transitivity implies that indifference curves cannot intersect.



$A \sim B$
 $A \sim D$

It doesn't matter whether
 ① $B > D$ OR ② $D > B$ we will
 get a contradiction

① $B \succ D$ but $A \sim B$, $A \sim D \Rightarrow B \sim D$!

② $D \succ B$, but $A \sim B$, $A \sim D \Rightarrow B \sim D$!

B and D are
on different
indifference curves!

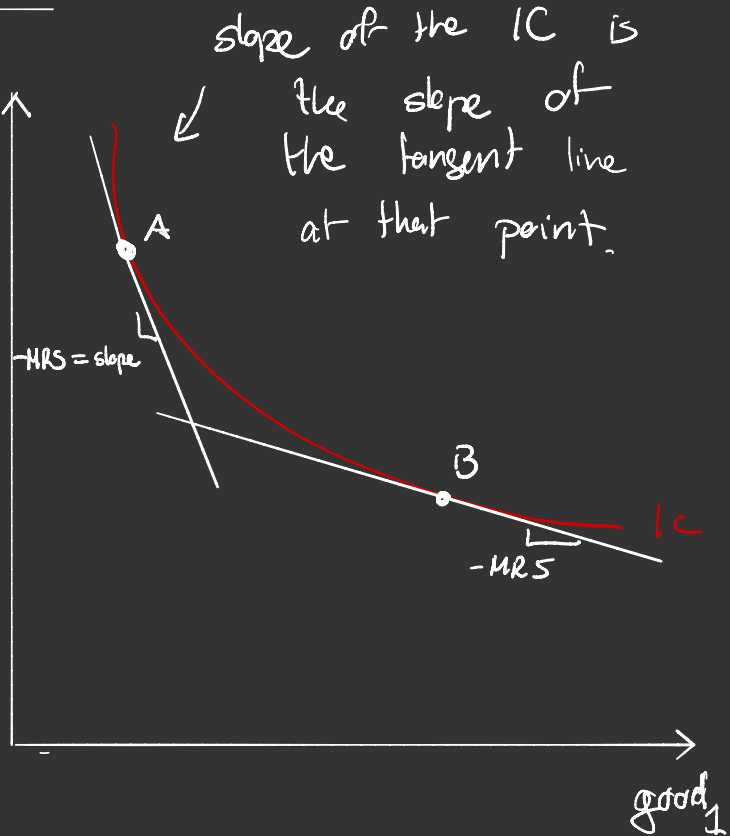
They can't be
 $B \sim D$!

The Marginal Rate of Substitution

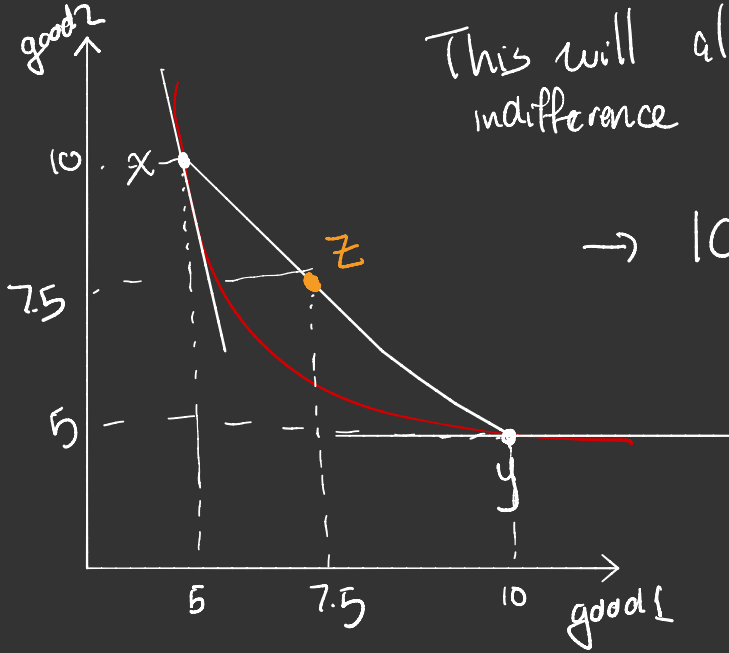
$MRS = | \text{slope of the indifference curve at a particular point} |$

$= \text{cost of opportunity of good 1 in terms of good 2}$

" Personal (in terms of consumer preferences) of remaining in the same indifference curve, measured as units of good 2 given up to get one more unit of 1 "



Convexity



This will always be the case for convex indifference curves: "preference for balanced consumption bundles"

→ IC is convex: bowed out shape

Note that $MRS_x > MRS_y$

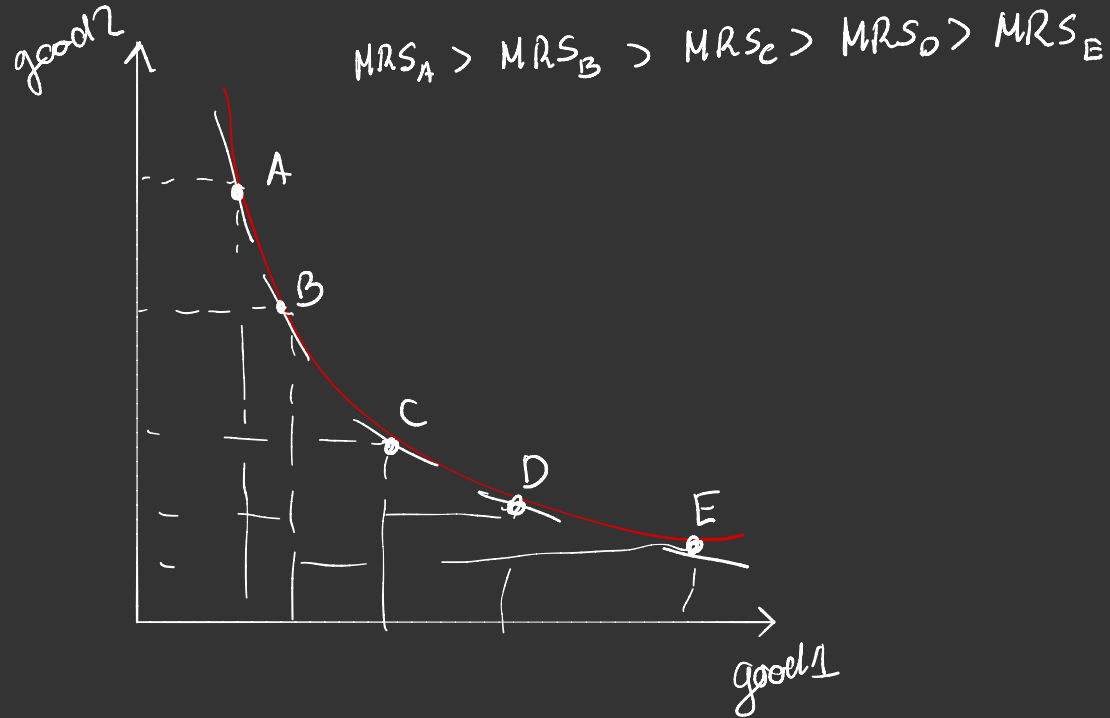
→ MRS is decreasing

$$x \sim y$$

z is a combination of x and y

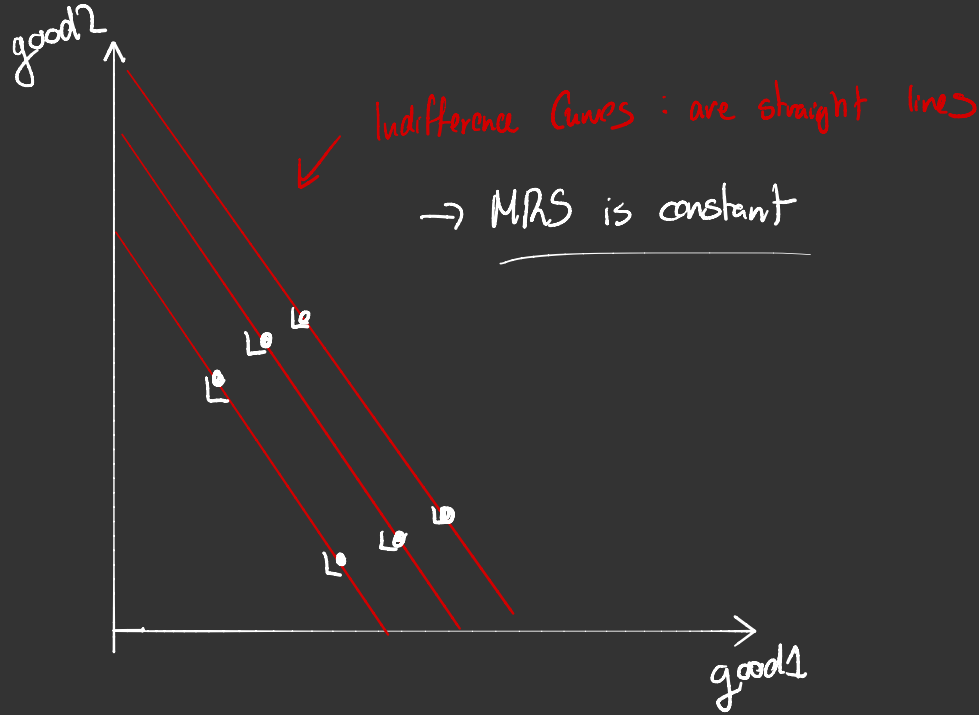
$z \succ x$, and $z \succ y$

In general, the MRS changes with the bundle in the IC we are considering



Perfect Substitutes

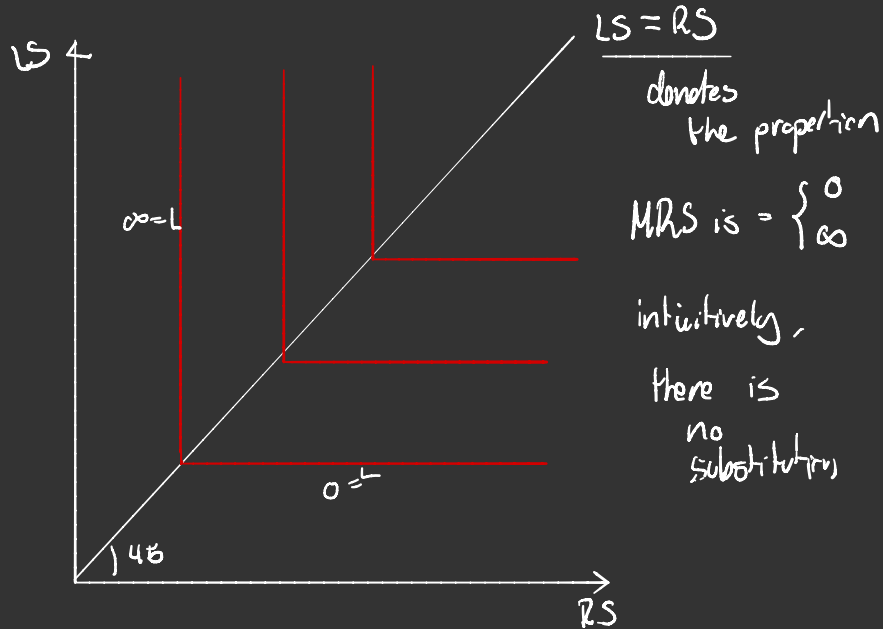
Preferences have constant MRS between them:



Perfect Complements: Goods need to be consumed together in a fixed proportion

Ex: Right and Left Shoes

good RS : LS
proportion 1 : 1

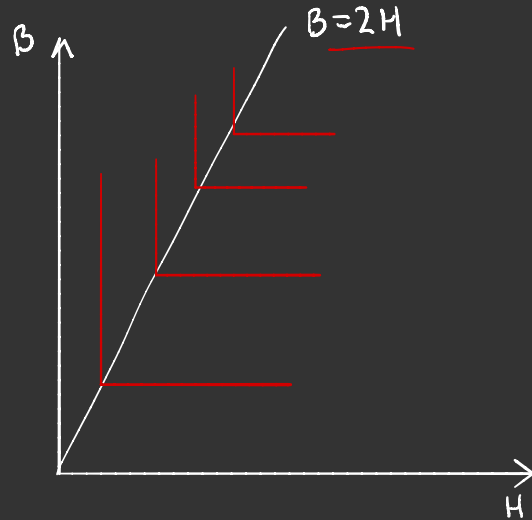


Ex. I like my sandwiches with

good (B)read : (H)am

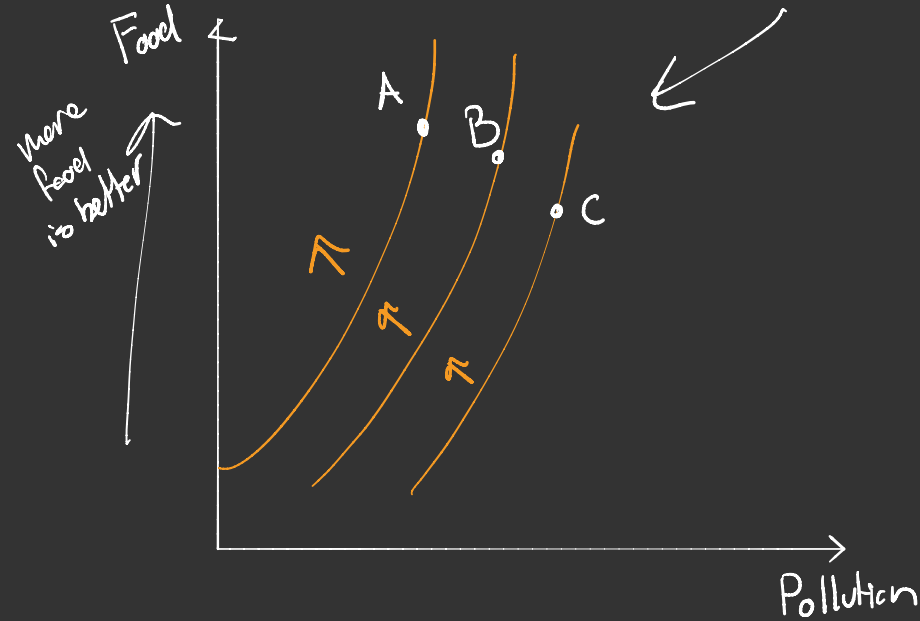
proportion 2 : 1 i.e. 2 slices of bread per 1 slice of ham.

$B = 2H$: for every slice of Ham, I need 2 slices of bread



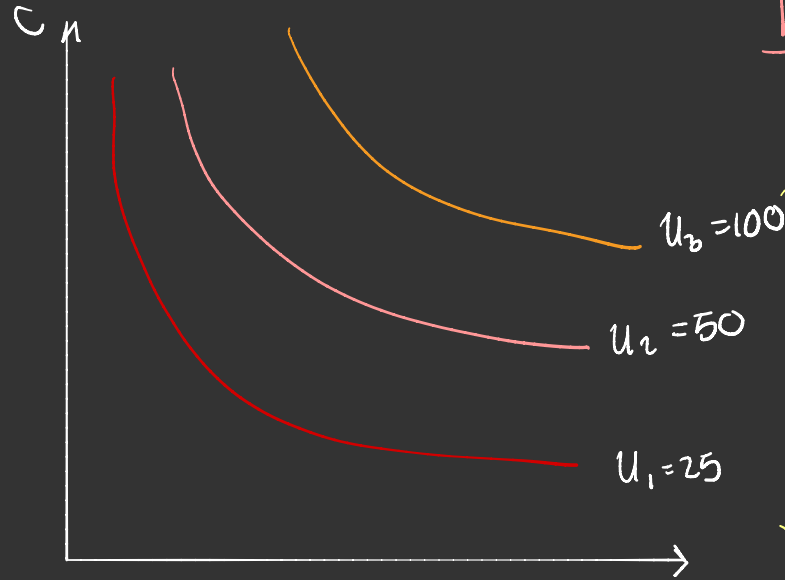
Bad : One good whose consumption I don't like, or, less is better

$$A > B > C$$



Less Pollution is better because it is a bad.

Ordinal vs Cardinal Utility



Indifference Curve map
comes from the Utility function

$$U(F, C) = F \cdot C \xrightarrow{f} 2U(F, C)$$

with cardinal utility, the number matters, because it represents a magnitude of How much we like the bundle.

Do not represent happiness/magnitude of preference, they just represent a ranking!

$\times 2$

$U_3 = 200$
 $U_2 = 100$
 $U_1 = 50$

Still represent the same ranking!

$\times 2$

Ordinal Utility
only ranking matters, not the number per se.

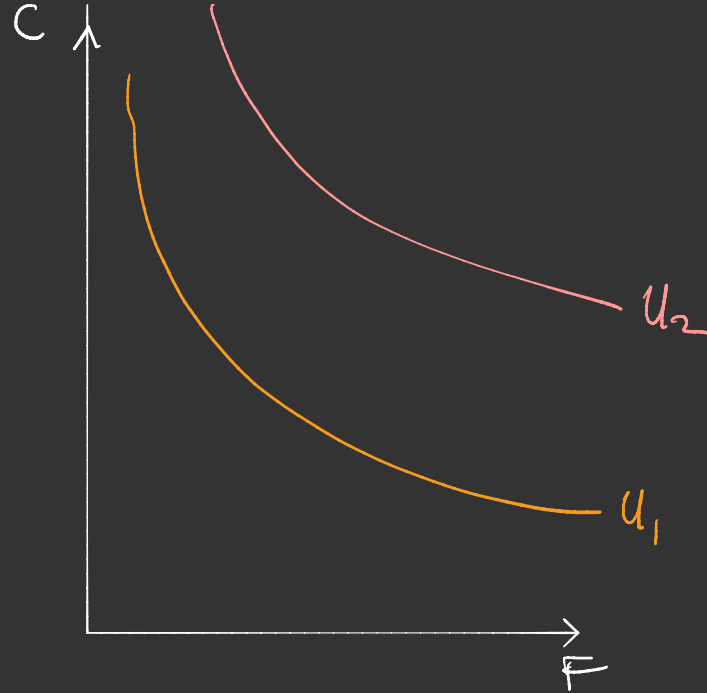
MOST of ECON (in this class)
we will focus on ordinal utility.

From Utility to Indifference Curves:

$U(F, C) = F \cdot C$, to compute the indifference curve, fix a level of utility and solve:

$\bar{U} = F \cdot C$: bundles that satisfy this equation give the same utility and thus are indifferent.

$C = \frac{\bar{U}}{F}$ eq. of the Indifference Curve with utility $\bar{U}$



Utility function Perfect Substitutes, has the following general

shape: $U(x_1, x_2) = a x_1 + b x_2$, $a, b > 0$

Ex.

$x_1 = (C) \text{ ore}$

$x_2 = (P) \text{epsi}$

$U(C, P) = 2C + P$ (Pepsi)

An indifference curve

$\bar{U} = 2C + P$

$P = \bar{U} - 2C$

$MRS = |\text{slope}|$
 $= |-2|$
 $= 2$



In general $\bar{U} = a x_1 + b x_2$

eq. of IC is $x_2 = \frac{\bar{U}}{b} - \frac{a}{b} x_1$; $MRS = \frac{a}{b}$

Perfect Complements: the general case can be describe by the utility function $u(x_1, x_2) = \min \{ax_1, bx_2\}$, $\begin{matrix} x_1 & : & x_2 \\ b & \swarrow \searrow & a \end{matrix}$

The "kink" of the right angles (IC's) is obtained by $bx_2 = ax_1$

$$\Rightarrow x_2 = \frac{a}{b} x_1; \quad \begin{matrix} x_1 = \text{Hum} \\ x_2 = \text{Bread} \end{matrix}$$

$$\begin{matrix} a=2 \\ b=1 \end{matrix} \Rightarrow u(H, B) = \min \{2H, B\}$$

"kink" $B = 2H$

$$MRS = \begin{cases} 0 \\ \infty \end{cases}$$

